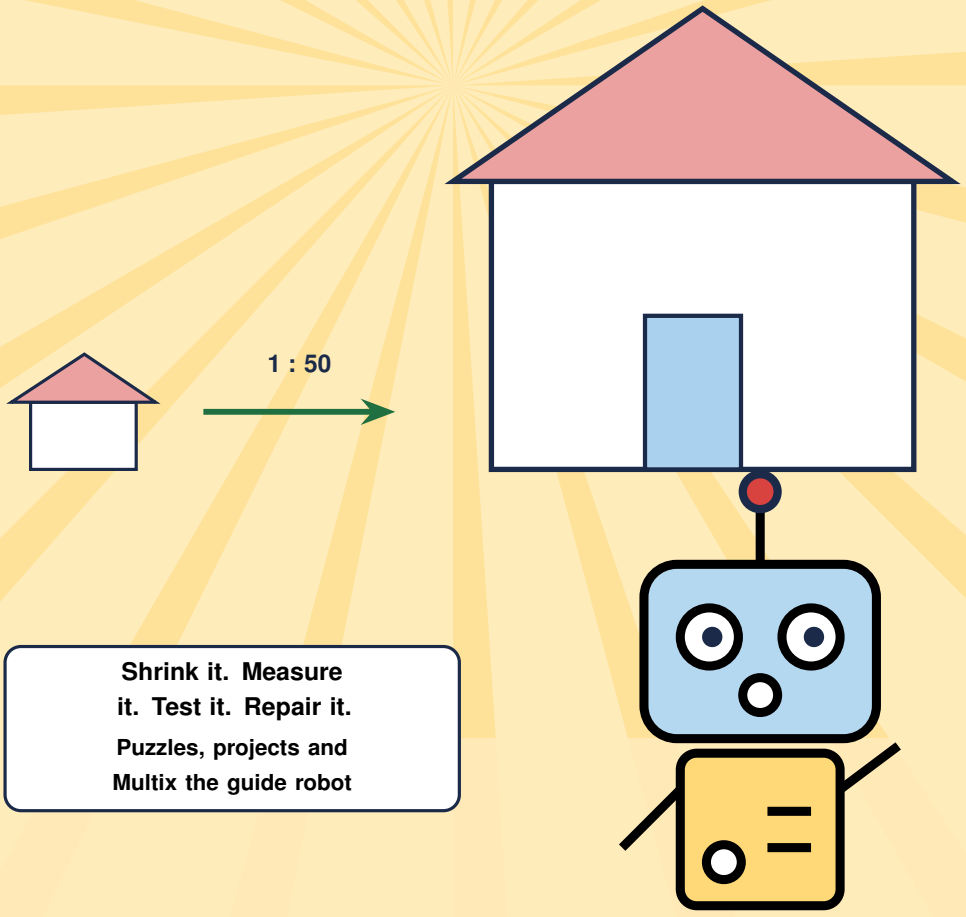
SCIENCE
Book 9

Small Models, Big Things



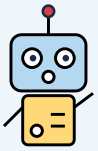
Shrink it. Measure it. Test it. Repair it.

Puzzles, projects and Multix the guide robot

Ages 9 to 11

This book belongs to _____

Hello. I am Multix.

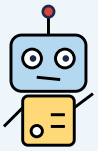


I am a guide robot. I read every instruction very carefully, which is why I sometimes get things wrong. When I make a mistake, see whether you can catch it. I never give away a puzzle answer.

How this book works

Every book in this series asks the same few questions, so once you have read this page you will know your way around.

- **Try it** boxes tell you what to do with a ruler, a cup, some string or a clock. You need nothing from a shop. Everything is at home or at school.
- **What did we keep?** After you simplify something, such as keeping the same conditions in a test or averaging several throws, you say what stayed and what was set aside. Simplifying is fine. Forgetting what you set aside is not.
- **What makes this correction trustworthy?** When you fix or improve an answer, a box says what you checked, what the check gives you reason to trust, and what it does *not* show.
- **Circles** show how a number was made. Page 4 explains how to read them, and Book 1 introduced them.
- **Puzzles** come at the end of each chapter. Try them before you look at the answers on page 45.
- **Slop check** is a page near the end. Slop is hasty, gaudy work that forgot to take care. You will learn to spot it and to fix it.



Here is a fact about me. I built a cardboard model of my room and announced that it was the room. Then I tried to sit on the model bed. Chapter 1 will explain why I should be embarrassed.

In this book

1	What a Model Keeps	5
2	Maps and Scales	10
3	Answering a Question	15
4	What Each Model Drops	20
5	Testing a Model	25
6	Repairing a Model	30
⋆	Slop check	37
⋆	Answers	45
⋆	Words, and notes for grown-ups	47

Try it: Gather your kit

You will use these things across the six chapters. Everything is found at home or at school. Measuring comes from Book 1, fair testing from Book 2 and the ledger from Books 5 to 8, and all of them come back here.

- a tape measure
- a ruler marked in mm
- plain and squared paper
- a pencil and an eraser
- cardboard and a box or two
- scissors (blunt ones are fine)
- sticky tape
- a notebook
- a room, a hallway or a street to measure
- a helper to hold the tape

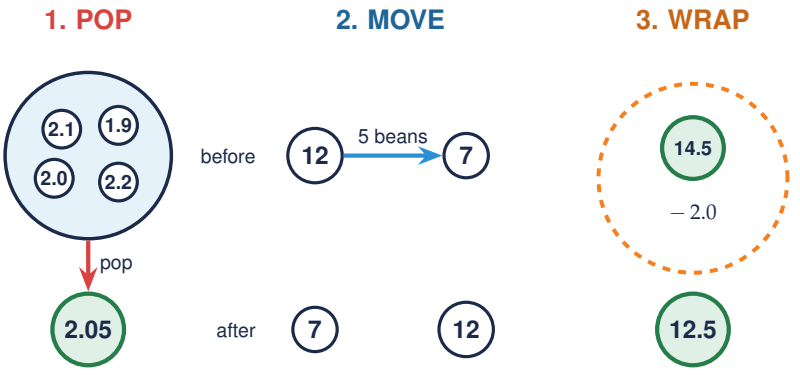
■ **Adult help:** Cut cardboard on a table with blunt scissors, and ask a grown-up to help with a craft knife if one is needed. Measure rooms and hallways that are safe to walk in, and stay away from roads, stairs and ladders. Nothing here needs a flame or mains electricity.

Units in this book

We use the metric system: millimetres (mm), centimetres (cm) and metres (m). A **scale** says how big the model is compared with the real thing. “1 cm stands for 50 cm” means every centimetre on the model is 50 cm in real life. We write it 1 : 50. There are 10 mm in 1 cm and 100 cm in 1 m. If you are used to inches, one inch is about 2.5 cm. We will use that only in side notes.

Reading the circles

A circle holds a group of things together. There are three things you can do with circles in this series.

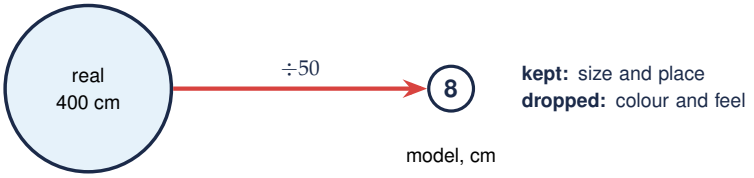


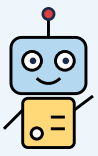
Pop Squash a group down to one value that sums it up, such as an average. A pop is a *summary*, so it always loses something. Every time we pop, we say what the summary keeps and what it loses.

Move An arrow that carries something from one circle to another, such as beans or water. Nothing is summarized. You will see more of moves in later books.

Wrap When you correct something, draw a new dashed circle around the old one and show what changed. The old work stays visible, so anyone can check it.

A model and its real thing





A pop is not a lie, but it is a shortcut. Any time I tell you a pop, I will tell you what it threw away.

1 What a Model Keeps

A small stand-in that shrinks every length by the same amount

A toy car, a cardboard house, a street map and a drawing of the heart are all **models**. A model is a small stand-in for something big, hard to reach or hard to see. It keeps some things about the real thing and drops the rest. That is why it is useful, and also why it can mislead.

A **scale model** keeps one thing carefully: every length is shrunk by the *same* number. If 1 cm on the model stands for 50 cm in the room, a 400 cm wall becomes $400 \div 50 = 8$ cm. We write the scale as 1 : 50.

Model	What it keeps	What it drops
A cardboard room	sizes and places	colour, texture, weight
A scale	the same shrink for every length	very small details
A measurement	a number to the nearest cm	the last few mm

Try it: Shrink a room

You need: a tape measure, a helper, a pencil and plain paper.

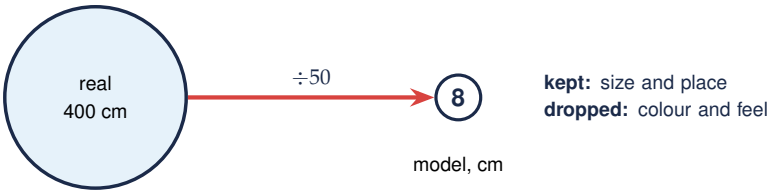
1. Pick a room. Measure its length and width, the width of a door and the length of the bed or the sofa.
2. Divide each real length by 50 to get the length on a 1 : 50 model.
3. Draw the room on paper using the model lengths.

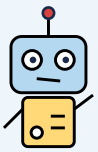
Thing	Real length (cm)	Model length (cm)
Room length		
Room width		
Door width		
Bed or sofa length		

■ **Adult help:** Stay clear of stairs and keep the tape low. Two people can hold the tape flat along the floor.

Practice numbers, not from a real room

Room length 400 cm, so the model is $400 \div 50 = 8$ cm. Room width 300 cm, so 6 cm. Door 80 cm, so 1.6 cm. Bed 200 cm, so 4 cm.
Check one backwards: $4 \times 50 = 200$.





The model room is 8 cm wide, so the real room is 8 cm wide. A model is the room, only smaller. I have been very sure about this. Am I right?

What makes this correction trustworthy?

What we checked. We measured each real length twice, divided each by the same number and checked one of them backwards.

What this check gives us reason to trust. Every length was shrunk by the same amount, so the shapes and the proportions of the room are kept.

What it does not show. Anything we left out, such as the colour of the walls or how the bed feels, and any question the model was not built to answer.

What did we keep?

We turned a room into a drawing. **We kept** _____ **and we set aside** _____.

Try it: Two scales

Shrink the same three things at two scales: 1 : 50 and 1 : 20. A smaller number after the colon means a bigger model.

Thing	Real (cm)	At 1:50	At 1:20
Bed			
Desk			
Door			

Practice numbers

Bed 200 cm: 4 cm at 1 : 50 and 10 cm at 1 : 20. Desk 120 cm: 2.4 cm and 6 cm. Door 80 cm: 1.6 cm and 4 cm.
The 1 : 20 model shows more detail but needs more paper. The 400 cm room would be 20 cm long at 1 : 20.

Puzzle 1.1: Shrink it

A sofa is 200 cm long. At 1 : 50 it is _____ cm long.

Puzzle 1.2: Grow it

A model rug is 3 cm by 2 cm at 1 : 50. The real rug is _____ cm by _____ cm.

Puzzle 1.3: Which is bigger?

Which model of the same bed is bigger, 1 : 20 or 1 : 50? _____ Why? _____

Puzzle 1.4: Kept or dropped?

Write K for kept or D for dropped in a 1 : 50 cardboard room.
(a) the length of the bed _____ (b) the colour of the bed _____ (c) where the door is _____
(d) how soft the bed is _____ (e) where the desk stands _____

Puzzle 1.5: Unfair model

Ana made the bed at 1 : 50 and the desk at 1 : 20. What goes wrong when she puts them in the same room? _____

Puzzle 1.6: More shrinking

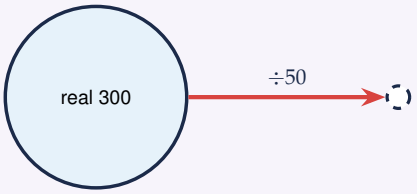
A chair is 50 cm wide and a table is 150 cm long. At 1 : 50 they are _____ cm and _____ cm.

Puzzle 1.7: Check a number

The model door is 1.6 cm wide at 1 : 50. How wide is the real door? _____ cm

Puzzle 1.8: Draw the circles

A wall is 300 cm long. At 1 : 50 it becomes _____ cm. Fill in the empty circle.



Puzzle 1.9: Wrap

A student wrote “The model is the room.” Wrap it with a clearer claim. _____

Puzzle 1.10: Design a model

Pick something at home to model, say your scale and say what you would keep. _____

Shrinking record

Shrink three things from a room of your choice. Use the scale shown in the first column.

Thing	Real (cm)	Scale	Model (cm)	Checked back?

Which scale was easier to draw on one page? _____

Which detail could your model not show? _____

Sketch your room at your chosen scale.

2 Maps and Scales

A place drawn flat, with a scale that says how small

A **map** is a model of a place drawn flat on paper. Its **scale** says how small it is. "1 cm to 1 m" means each centimetre on the map is one metre on the ground. A **scale bar** is a short ruler drawn on the map, so the scale still works if the map is copied bigger or smaller.

A map keeps places and distances. It drops anything that will not fit or does not help, such as the pictures on the wall, and anything too small to draw.

Model	What it keeps	What it drops
Map, 1 cm to 1 m	door places and lengths	pictures, small objects
Map, 1 cm to 2 m	order and long lengths	anything under 1 m wide
A scale bar	the scale, even if copied	nothing

Try it: Map a hallway

You need: a tape measure, squared paper, a pencil and a ruler.

1. Measure a hallway or a path in metres: its length, its width and where each door sits from the left end.
2. Draw it at 1 cm to 1 m. Then draw it again at 1 cm to 2 m.
3. Add a scale bar to each map.

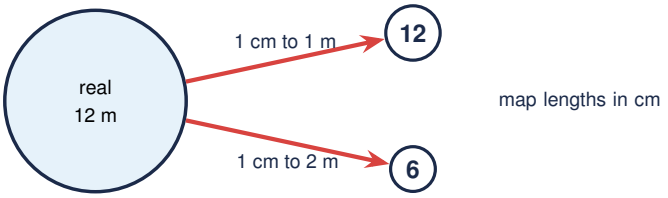
Thing	Real (m)	1 cm to 1 m (cm)	1 cm to 2 m (cm)
Length			
Width			
First door			

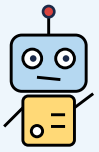
■ **Adult help:** Choose a hallway or a path that is safe to walk. Stay away from roads, stairs and ladders.

Practice numbers, not from a real hallway

A hallway is 12 m long and 2 m wide. Its doors are 1 m wide, with their left edges at 2 m, 6 m and 10 m.
At 1 cm to 1 m: 12 cm by 2 cm, doors 1 cm wide at 2, 6 and 10 cm. At 1 cm to 2 m: 6 cm by 1 cm, doors 0.5 cm wide at 1, 3 and 5 cm.

A 50 cm window is 0.5 cm at the first scale and 0.25 cm at the second. That is thinner than a pencil line, so the second map drops it.





The map at 1 cm to 2 m is half the size and fits on the page, so it is the better map. Smaller always wins. Do you agree?

What makes this correction trustworthy?

What we checked. We measured the hallway twice, used one scale for every length on each map and checked a length by measuring it back.

What this check gives us reason to trust. Measuring the map and multiplying gave the real length again, so the scales were applied the same way everywhere.

What it does not show. Anything thinner than the pencil line, and anything we chose not to draw. A map says nothing about what it dropped.

What did we keep?

We turned a hallway into two maps. **We kept** _____ **and we set aside** _____.

Try it: Read the map

Use your 1 cm to 2 m map. Measure the gap between the first and third door on the map and multiply by 2 for metres.

Gap	On the map (cm)	Real (m)
First door to third door		

Practice numbers

Doors at 1 cm and 5 cm on the map. Gap $5 - 1 = 4$ cm, so $4 \times 2 = 8$ m.
Check on the real hallway: doors at 2 m and 10 m, a gap of $10 - 2 = 8$ m.

Puzzle 2.1: Map to real

At 1 cm to 2 m, 3 cm on the map is _____ m on the ground.

Puzzle 2.2: Real to map

At 1 cm to 4 m, a real distance of 8 m is _____ cm on the map.

Puzzle 2.3: Too small

A window is 50 cm wide. At 1 cm to 1 m it is _____ cm. At 1 cm to 2 m it is _____ cm.

Puzzle 2.4: Pick a scale

You want a poster-sized map of one classroom. Which fits better, 1 cm to 10 cm or 1 cm to 10 m? _____

Puzzle 2.5: Detail to keep

A fire-drill map must keep some things. Write K for must keep or D for can drop.
(a) exit doors _____ (b) colour of the carpet _____ (c) stair places _____ (d) pictures on walls _____

Puzzle 2.6: Scale bar

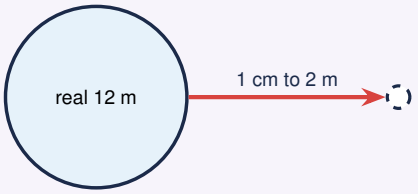
A map's scale bar is 2 cm long and stands for 4 m. A copier makes the map 1.5 times bigger. How long is the bar now? _____ cm Does it still tell the truth? _____

Puzzle 2.7: Unfair claim

Ana's map of the hallway at 1 cm to 1 m is 12 cm long. Ben's at 1 cm to 2 m is 6 cm long. Ana says her hallway is longer. What is wrong? _____

Puzzle 2.8: Draw the circles

A hallway is 12 m long. Its map at 1 cm to 2 m is _____ cm long. Fill in the empty circle.



Puzzle 2.9: Wrap

A student wrote "The map shows everything in the hallway." Wrap it with a better claim.

_____

Puzzle 2.10: Design a path map

Pick a path from your door to a place you visit. Say your scale and what you would have to drop. _____

Map record

Map three paths at the scales shown. Write the real length first.

Path	Real (m)	Scale	Map (cm)	Checked back?

Which scale fit best on your page? _____

What had to be dropped to make it fit? _____

Sketch one map and add its scale bar.

3 Answering a Question

A model is built for some questions and not others

Every model is good at some questions and useless at others. A map can say which way to the library but not what the library smells like. A scale model can say whether a wardrobe fits an alcove but not how the wardrobe feels. Before using a model, ask: **what must this model keep to answer my question?**

Question	A model that suits it	What it must keep
Which way to the library?	a map	places and routes
Will the wardrobe fit the alcove?	a scale model	sizes
How does blood flow round a heart?	a diagram	the order of the parts

Try it: Ask the model

Use your 1 : 50 room model from chapter 1. Write three questions about the room and predict each answer from the model alone. Then check the real room.

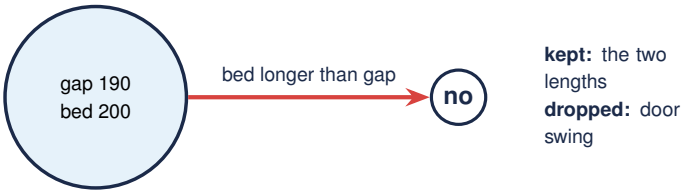
Question	Model says	Real room	Match?

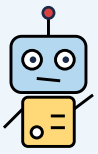
Practice numbers, not from a real room

Q1. Does the 200 cm bed fit between the desk and the end wall? The model gap is 3.8 cm, so $3.8 \times 50 = 190$ cm. The bed is 4 cm on the model. Model says: no. Real gap 190 cm. Bed 200 cm. No.

Q2. Does the 120 cm desk fit under the 150 cm window? Model: 2.4 cm in 3.0 cm. Model says: yes. Real: yes.

Q3. Does the door open fully? The model drew the door as a line with no swing. The model cannot say.





The model is of the whole room, so it can answer every question about the room, including whether the door opens. I like a model that knows everything. Does it?

What makes this correction trustworthy?

What we checked. We predicted two answers from the model before measuring, then measured the real room and compared.

What this check gives us reason to trust. Both predictions matched, so for questions about sizes and gaps this model has earned some trust.

What it does not show. Anything the model did not keep, such as door swing, light or colour, and any gap that is much smaller than our measuring error.

What did we keep?

We turned questions into predictions. **We kept** _____ **and we set aside** _____.

Puzzle 3.1: Pick the model

Write M for map, S for scale model or D for diagram.
(a) which way to the library _____ (b) will the wardrobe fit the alcove _____ (c) how blood flows round a heart _____

Puzzle 3.2: Read the gap

The model gap between desk and wall is 3.8 cm at 1 : 50. The real gap is _____ cm. The bed is 200 cm. Does it fit? _____

Puzzle 3.3: Spare room

Desk 120 cm, window 150 cm. Spare space _____ cm.

Puzzle 3.4: Not built for it

Can the room model say what colour the carpet is? _____ Why not? _____

Puzzle 3.5: Unfair check

Ben tested the room model against a different room and says it is wrong. What is the problem? _____

Puzzle 3.6: Chair fit

A model chair is 1.2 cm wide at 1 : 50. The real chair is _____ cm wide. A model gap is 1.0 cm, so the real gap is _____ cm. Does the chair fit? _____

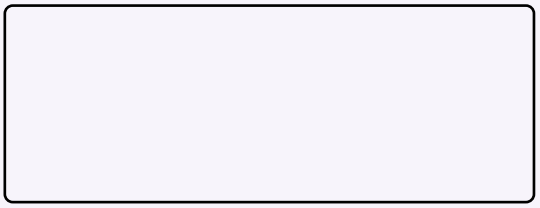
Puzzle 3.7: A question it cannot answer

Name one question your room model cannot answer.

_____

Puzzle 3.8: Draw the circles

Real gap 190 cm and bed 200 cm go into a question circle. Draw the answer circle.



Puzzle 3.9: Wrap

A student wrote “The model says everything is fine.” Wrap it with a better claim.

_____

Puzzle 3.10: Design a test

Say how you would find out whether your bike fits in the shed, using a model.

_____

Question record

Write three more questions. Predict from the model, then check the real thing.

Question	Model says	Real thing	Match?

Which question could the model not answer? _____

What would the model need to keep to answer it? _____

Sketch the part of your model that answers your best question.

4 What Each Model Drops

Two diagrams of one thing, two sets of answers

The same thing can have several models, and each one drops something different. A bus route has stops in a certain order, and the stops are some distance apart. A street map keeps the distances. A route diagram, the kind printed inside a bus, keeps only the order of the stops and spaces them evenly. Both are true to what they keep, and each can mislead about what it dropped.

Model	What it keeps	What it drops
Street map to scale	distances and shape	the number of stops
Route diagram	order of stops	real distances
A model of any kind	what its maker needed	what they did not

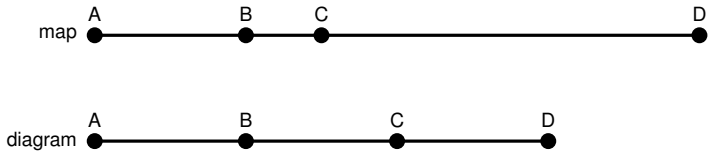
Try it: Two diagrams of one walk

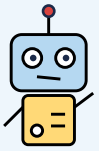
Pick a walk, such as from home to a shop, with at least three stops or landmarks. Draw it twice: once as a map with the distances to scale, and once as a route diagram with the stops evenly spaced. Then ask both diagrams the same three questions.

Question	Map says	Diagram says
Which two stops are closest?		
How many stops after the first?		
How far is it in all?		

Practice numbers, not from a real route

Four stops A, B, C, D. Real gaps: A to B 200 m, B to C 100 m, C to D 500 m. Total $200 + 100 + 500 = 800$ m.
Map at 1 cm to 100 m: gaps of 2, 1 and 5 cm, 8 cm in all. Route diagram: every gap drawn as 2 cm, 6 cm in all.
The map shows B and C are closest. The diagram shows three equal gaps and cannot say so.





The diagram shows the stops evenly spaced, so they are all the same distance apart. I trust drawings very much. Should I?

What makes this correction trustworthy?

What we checked. We asked both diagrams the same three questions and compared each answer with the real distances.

What this check gives us reason to trust. It showed which question each diagram can answer: the map for distance, the diagram for order and count.

What it does not show. Anything neither diagram kept, such as hills, traffic or how long the walk takes.

What did we keep?

We turned one route into two diagrams. **We kept** _____ **and we set aside** _____.

Puzzle 4.1: Add up

Real gaps are 200, 100 and 500 m. Total _____ m. At 1 cm to 100 m the map is _____ cm long.

Puzzle 4.2: Which diagram?

Write M for the map, R for the route diagram or X if neither can answer.
(a) how far from A to D _____ (b) how many stops after A _____ (c) are there steep hills _____

Puzzle 4.3: Equal gaps?

On the route diagram C to D looks as far as B to C. In real life it is how many times farther?
_____

Puzzle 4.4: Kept and dropped

List one thing the route diagram keeps and one it drops.
_____

Puzzle 4.5: Model or real

Write M for model and R for real thing.
(a) the route diagram _____ (b) the street _____ (c) the map _____

Puzzle 4.6: A tourist on foot

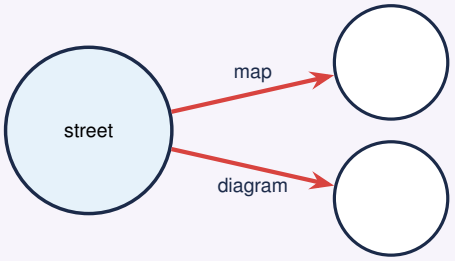
Which would you give a tourist who must walk, the map or the diagram? _____
Why? _____

Puzzle 4.7: A bus passenger

Which would you give a passenger who counts stops? _____ Why? _____

Puzzle 4.8: Draw the circles

One street goes into two circles, a map and a diagram. Write one thing each keeps.



Puzzle 4.9: Wrap

A student wrote "The diagram is the better model." Wrap it with a better claim. _____

Puzzle 4.10: Design two

Choose something with parts, like a school or a body. Say what two diagrams of it would each drop. _____

Two diagrams

Draw your own walk both ways. Label each stop.

Map, drawn to scale

Route diagram, stops evenly spaced

Which question did the map answer that the diagram could not? _____

Which question did the diagram answer more easily? _____

5 Testing a Model

Comparing what it predicts with the real thing

A model earns trust by being tested. Ask it a question, write the answer *before* measuring, then measure the real thing and compare. The difference is a **gap**. A small gap is good, but how small is small enough depends on the question.

Model	What it keeps	What it drops
A prediction	what the model says	how sure we should be
A gap	how far off the model was	why it was off
A margin	how much room a question has	the model's own errors

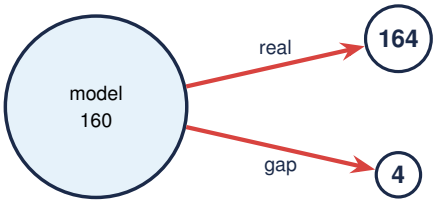
Try it: Predict, then check

Use your 1 : 50 model. Choose three distances in the room. For each, measure it on the model, multiply by 50 for a prediction, and then measure the real distance. Work out the gap.

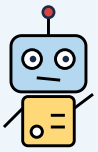
Distance	Model says (cm)	Real (cm)	Gap (cm)

Practice numbers, not from a real room

Bed to door: model 3.2 cm, so 160 cm. Real 164 cm. Gap 4.
Bed length: model 4 cm, so 200 cm. Real 200 cm. Gap 0.
Desk end to wall: model 1.2 cm, so 60 cm. Real 58 cm. Gap 2.
The biggest gap is 4 cm.



$164 - 160 = 4$



The gaps are 4, 0 and 2 cm, which are tiny. So the model is correct and good for every question. I will now use it to fit a chair 59 cm wide into a gap of 60. Safe?

What makes this correction trustworthy?

What we checked. We wrote each prediction before measuring, then measured the real room and worked out every gap.

What this check gives us reason to trust. The three gaps were all 4 cm or less, so the model can answer questions with a margin of more than about 5 cm.

What it does not show. Questions with a smaller margin. A 4 cm gap is nothing for a bed and a lot for a chair squeezed into a narrow gap.

What did we keep?

We turned a model into predictions and gaps. **We kept** _____ **and we set aside** _____.

Try it: Check the margin

A **margin** is the spare room a question has. Pick things to fit into gaps in your room and write the margin for each. A model can be trusted for a question when the margin is bigger than its biggest gap.

Thing in gap	Thing (cm)	Gap (cm)	Margin	Safe?

Practice numbers

Basket 40 cm in a 60 cm gap: margin 20, bigger than 4, safe.
Chair 59 cm in a 60 cm gap: margin 1, smaller than 4, not safe.
Bed 200 cm in a 190 cm gap: margin −10. The bed does not fit, and 10 is bigger than 4, so the answer “no” is safe.

Puzzle 5.1: Find the gap

Model says 160 cm and real is 164 cm. The gap is _____ cm.

Puzzle 5.2: Predict

A model distance is 2.6 cm at 1 : 50. Predict the real distance. _____ cm

Puzzle 5.3: Margin

A basket is 40 cm and the gap is 60 cm. Margin _____ cm. Safe if the biggest gap is 4 cm? _____

Puzzle 5.4: Tight fit

A chair is 59 cm and the gap is 60 cm. Margin _____ cm. Safe? _____

Puzzle 5.5: Safe “no”

A bed is 200 cm and the gap is 190 cm. The model says it does not fit. Is that answer safe? _____ Why? _____

Puzzle 5.6: Why three?

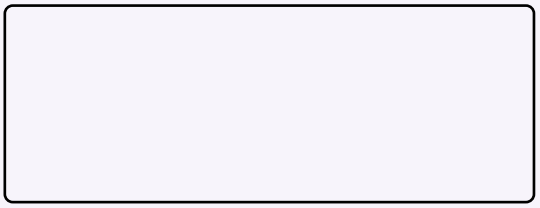
Why test a model on three predictions and not just one? _____

Puzzle 5.7: Unfair test

Ana measured the model with a ruler and the real room with a tape, then wrote down only the one gap that was zero. What is wrong? _____

Puzzle 5.8: Draw the circles

Model says 60, real is 58. Draw the circles for the model, the real value and the gap.



Puzzle 5.9: Wrap

A student wrote “The model is correct.” Wrap it with a better claim. _____

Puzzle 5.10: Design a test

Plan a test of your own model and say which question you will ask it. _____

Prediction record

Test your model on five predictions.

Distance	Model says	Real	Gap	Margin of question

What is your biggest gap? _____

For what size of question can you trust your model? _____

Draw your gaps as a bar chart.

6 Repairing a Model

When a model is wrong, find out why and wrap it

A model that gives a wrong prediction is not useless. The job is to find *why* it was wrong, then repair it. Often the model is fine for big questions and fails only for a small one. The repair is a better model for that small question, and the old model stays as it was, wrapped inside the new account.

Model	What it keeps	What it drops
A 1 : 50 model	rooms and big gaps	anything under 5 cm
A 1 : 10 model	small gaps in one corner	the rest of the room
A repair	the old model and the fix	the claim it replaced

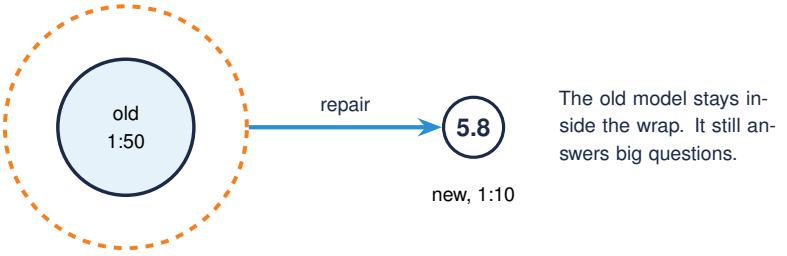
Try it: Repair it

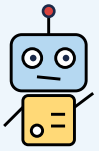
Find the question in chapter 5 with the smallest margin. Rebuild just that corner at 1 : 10 and ask the question again.

Scale	Model gap (cm)	Model thing (cm)	Model says	Real
1 : 50				
1 : 10				

Practice numbers, not from a real room

A chair is 59 cm wide. The real gap is 58 cm.
At 1 : 50: gap $58 \div 50 = 1.16$, chair $59 \div 50 = 1.18$. A ruler marked in mm reads both as 1.2 cm, so the model says “fits”. One mm on the model is 5 cm of room, too coarse.
At 1 : 10: gap 5.8 cm, chair 5.9 cm. The chair is bigger. The model now says “does not fit”.
Real: it does not fit.





The model said the chair fits, but it does not, so the model is broken. I will throw it in the bin and start again with nothing. Is that wise?

What makes this correction trustworthy?

What we checked. We repeated the same question with the 1 : 10 corner model and compared the answer with the real chair.

What this check gives us reason to trust. At 1 : 10 one mm is only 1 cm of room, which is finer than the 1 cm margin, so the new answer is trustworthy for this question.

What it does not show. Other corners. We have only checked this one at 1 : 10, and the new model says nothing about the rest of the room.

What did we keep?

We repaired a model for one question. **We kept** _____ **and we set aside** _____.

Spot the map that cheats

All three maps show the same hallway, 6 m long, with a door that is 1 m wide. Two of them are drawn correctly. One is not.

A scale: 1 cm to 1 m



B 1 cm to 2 m



C scale: 1 cm to 2 m



Which map cheats? _____ How wide would its door be in real life? _____ m

Which map shows the door at its correct width with the smaller scale? _____

Puzzle 6.1: Reading precision

At 1 : 50, 1 mm on the model stands for _____ cm of room.

Puzzle 6.2: Finer model

At 1 : 10, 1 mm on the model stands for _____ cm of room.

Puzzle 6.3: Re-ask

At 1 : 10 the chair is 5.9 cm and the gap is 5.8 cm. Does the chair fit? _____

Puzzle 6.4: Which first?

A model said a 90 cm shelf fits a 95 cm gap, but the real gap is 88 cm. Name the first thing you would check. _____

Puzzle 6.5: Bin or repair?

Why is it better to repair a model than to throw it away? _____

Puzzle 6.6: Pick the fix

Choose a fix: (a) remeasure (b) a bigger scale for that corner (c) ask a different question. A 1 : 50 model reads a 1 cm margin wrongly. _____

Puzzle 6.7: Model or real?

Write M for model and R for real thing.
(a) the 1 : 10 corner _____ (b) the chair _____ (c) the old 1 : 50 room _____

Puzzle 6.8: Which question?

Can the repaired corner model answer "does the door open fully?" _____ What would it need? _____

Puzzle 6.9: Unfair repair

Ben made the corner at 1 : 10 but measured the chair at 1 : 20. Why does that mislead?
_____

Puzzle 6.10: Check backwards

At 1 : 10 the chair is 5.9 cm and the gap 5.8 cm. The real sizes are _____ cm and _____ cm. The difference is _____ cm.

Puzzle 6.11: Wrap

A student wrote “The model was wrong.” Wrap it with a better claim. _____

Puzzle 6.12: A good habit

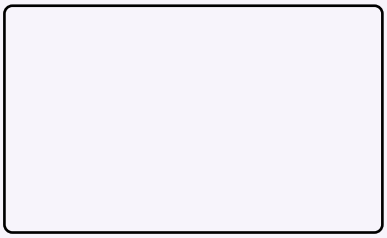
Write one habit from this book that makes a model more trustworthy. _____

Puzzle 6.13: Missing step

A student built a model and said it was good without testing it once. What could go wrong? _____

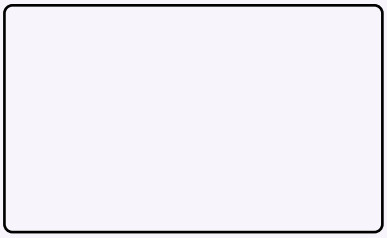
Puzzle 6.14: Draw the pair

Draw the real chair and its model as two circles joined by a rule.



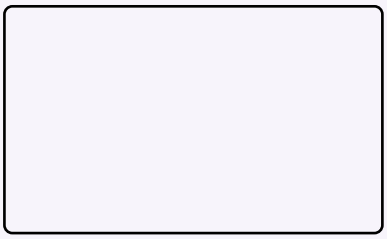
Puzzle 6.15: Draw the repair

Draw the 1 : 10 corner with the old 1 : 50 room still inside a dashed circle.



Puzzle 6.16: Draw the wrap

Draw a circle around “the model was wrong” and write the corrected claim outside it.

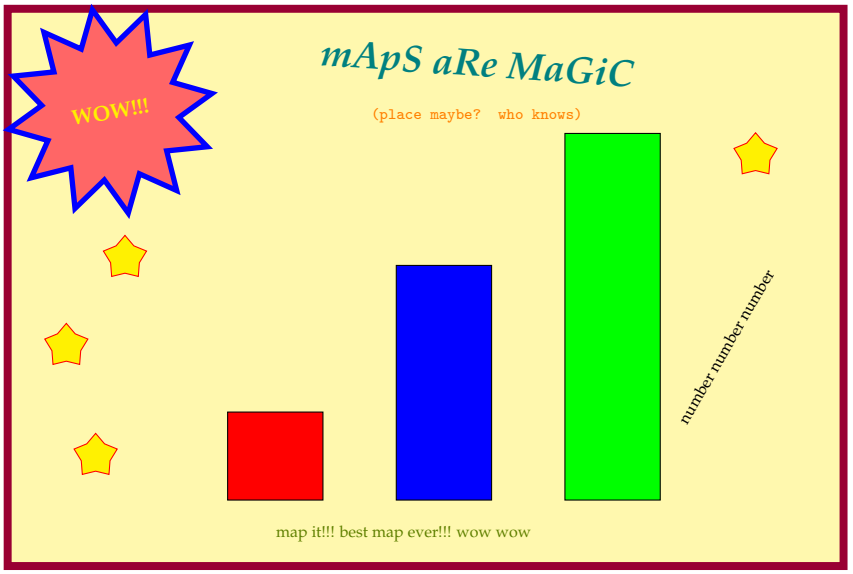


Slop check

Slop is hasty, gaudy work that forgot to take care. It can be made by a person or a machine, quickly or slowly. What matters is whether the work shows care for three things:

- **Salient:** Is the main point easy to see?
- **Simple:** Is there anything that does nothing but get in the way?
- **Relevant:** Does everything on the page help answer the question?

Here is a hasty poster about a hallway map. Study it, then decide what is wrong.



Slop check

(a) What is the poster's main point? Can you tell?

_____

(b) Name two things that are not simple, because they only distract.

_____

(c) Name two things that are missing, so the picture cannot be read.

_____

(d) Which part is not relevant to a hallway at all?

_____

Now you try

On a separate sheet, redraw a clear map of a hallway: a hallway 12 m long and 2 m wide with doors at 2, 6 and 10 m, drawn at 1 cm to 1 m. Give it a title, add a scale bar, label the doors, use one or two calm colours, and delete everything that does not help. Check it against the three words: salient, simple, relevant.



Scale bar:

Title:

Build and test your own model

Choose something real, such as a room, a desk, a path or a shelf. Write one question the model must answer, then build it, test it and say what it cannot do.

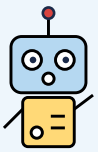
The real thing	
My question	
My scale	
What my model keeps	
What my model drops	
My prediction	
The real answer	
The gap, and the margin	
Questions it cannot answer	

Sketch the model and mark its scale.

What did we keep, across the whole course?

Every book in Science built a model. For each book, write what the model kept and what it dropped. Book 1 is done for you. Use the hints in the second column.

Book	Stood in for	What we kept	What we set aside
1 The Ruler Is Never Exact	a length by a reading	the number and its error	the true length
2 A Fair Test	a whole by a trial		
3 Shadows That Tell Time	a time by a shadow		
4 Things That Push Back	a force by an arrow		
5 Where the Water Goes	water by a count		
6 Mixed Up and Sorted Out	a mixture by weights		
7 Sorting the Living	a group by a key		
8 Where the Energy Goes	energy by a height		
9 Small Models, Big Things	a thing by a model		



Every model in this book was a stand-in. A stand-in is allowed to drop things, as long as you say which and test it against the real thing. Whenever you meet a map, a chart or a diagram, you will know to ask what it kept, what it dropped and which questions it was never built for.

Write a model report

1. **What I looked at.**
2. **What I changed, measured and kept the same.**
3. **My readings.** All of them.
4. **My answer.** With the gap named.
5. **What I kept and what it hides.**

Pick one model from this book and write a report on it. Use the lines below.

Tests to try

Pick one and design it with the habits from this book: change one thing, keep the rest the same, repeat, and test the model against the real thing.

1. Does a 1:20 model give smaller gaps than a 1:50 model?
2. Which is easier to follow, a map or a route diagram, for a stranger?
3. How closely can a model of your desk predict where a book will fit?
4. How much detail can a 1 cm to 5 m map of your street keep?
5. How wide must a gap be before a 1:50 model is safe?
6. Can a model of a shoebox predict how many fit on a shelf?

Write the test you pick:

_____

_____

_____

_____

_____

_____

Change: _____ Measure: _____ Keep the same: _____

A week of models

Each day, notice one model, such as a map, a chart or a toy, and say what it keeps and drops.

Day	The model	What it drops
Mon		
Tue		
Wed		
Thu		
Fri		
Sat		
Sun		

Questions I still have

Write what you still wonder about models. Then pick one and think how to test it.

Answers

Try every puzzle first. Where an answer shows a calculation, check the steps, not just the result.

Chapter 1

Multix on the 8 cm. No. 8 cm on the model stands for 400 cm in the room, because $8 \times 50 = 400$. A model is a stand-in, not the room. **Keep box.** Kept: size and place. Set aside: colour and how things feel.

1.1 $200 \div 50 = 4$ cm.
1.2 $3 \times 50 = 150$ and $2 \times 50 = 100$. So 150 cm by 100 cm.
1.3 1 : 20. Each model centimetre stands for fewer real centimetres, so the model is bigger.
1.4 (a) K (b) D (c) K (d) D (e) K.
1.5 The bed and desk are at different scales, so their sizes do not match their real sizes.
1.6 Chair 1 cm. Table 3 cm.
1.7 $1.6 \times 50 = 80$ cm.
1.8 $300 \div 50 = 6$.
1.9 "The model is a smaller stand-in that keeps the room's sizes and places."
1.10 Answers vary. Check that the scale is stated and that a kept and a dropped thing are named.

Chapter 2

Multix on the half-size map. No. Smaller fits the page but it drops more. A 50 cm window is 0.25 cm at 1 cm to 2 m, thinner than a pencil line. **Keep box.** Kept: door places and long lengths. Set aside: pictures and small things.

2.1 $3 \times 2 = 6$ m.
2.2 $8 \div 4 = 2$ cm.
2.3 0.5 cm and 0.25 cm.
2.4 1 cm to 10 cm. A room is a few metres, which fits that scale on a poster.
2.5 (a) K (b) D (c) K (d) D.
2.6 $2 \times 1.5 = 3$ cm. Yes, the bar was copied with the map, so it still says 3 cm is 4 m.
2.7 The maps use different scales, so the lengths cannot be compared without the scales. Both hallways are 12 m.
2.8 6.
2.9 "The map keeps door places and lengths, and drops small things such as pictures."
2.10 Answers vary. Check the scale and at least one dropped thing.

Chapter 3

Multix on every question. No. A model answers only the questions its kept parts bear on. The door swing was not kept. **Keep box.** Kept: the question and the prediction. Set aside: anything the model was not built for.

3.1 (a) M (b) S (c) D.
3.2 $3.8 \times 50 = 190$ cm. No, the bed is longer than the gap.
3.3 $150 - 120 = 30$ cm.
3.4 No. Colour was not kept in a cardboard model.
3.5 A model of one room cannot be tested against a different room.
3.6 Chair $1.2 \times 50 = 60$ cm. Gap $1.0 \times 50 = 50$ cm. No, the chair is 10 cm too wide.
3.7 Answers vary, for example whether the door opens fully.
3.8 The answer circle says "no".
3.9 "The model says the bed does not fit, and it kept sizes but not door swing."
3.10 Cut a bike-sized outline at a scale and compare it with a shed outline at the same scale.

Chapter 4

Multix on equal gaps. No. The diagram keeps order and drops distance, so equal spacing means nothing about real distance. **Keep box.** Kept: order of stops, or distances. Set aside: distances, or the number of stops.

4.1 $200 + 100 + 500 = 800$ m. $800 \div 100 = 8$ cm.
4.2 (a) M (b) R (c) X.
4.3 $500 \div 100 = 5$ times farther.
4.4 Keeps order and count of stops. Drops distances.
4.5 (a) M (b) R (c) M.
4.6 The map, because walking depends on distance.
4.7 The diagram, because counting stops depends on order.
4.8 Map keeps distances. Diagram keeps order.
4.9 "Each diagram is better for a different question."
4.10 Answers vary. Check that each diagram has a kept and a dropped thing.

Chapter 5

Multix on the 59 cm chair. No. A 59 cm chair in a 60 cm gap has a margin of 1 cm, smaller than the model's biggest gap of 4 cm. **Keep box.** Kept: the prediction and the gap. Set aside: why it was off.

5.1 $164 - 160 = 4$.
5.2 $2.6 \times 50 = 130$ cm.
5.3 $60 - 40 = 20$. Yes, 20 is bigger than 4.
5.4 $60 - 59 = 1$. No, 1 is smaller than 4.
5.5 Yes. The bed is 10 cm too long, and 10 is bigger than 4.
5.6 One prediction could be lucky. Three show how big the gaps usually are.
5.7 Different tools and cherry-picking one zero gap hide the real spread. Use one method and report every gap.
5.8 Circles: 60, 58 and gap 2.
5.9 "The model's predictions were within 4 cm in the three tests we made."
5.10 Answers vary. Check that a prediction is written before measuring.

Chapter 6

Multix on the bin. No. The model is fine for large questions. Repair it for the small one and keep both, with the old model wrapped. **Keep box.** Kept: the old model and the fix. Set aside: the claim it replaced.

Spot the cheat. B. At 1 cm to 2 m its door is 1 cm, which would be 2 m wide. A and C show a 1 m door.
6.1 1 mm stands for 5 cm.
6.2 1 cm.
6.3 No. The chair is bigger than the gap.
6.4 Remeasure the real gap and the model, then see whether a bigger scale is needed.
6.5 The old model still answers big questions, and it shows what was tried.
6.6 (b).
6.7 (a) M (b) R (c) M.
6.8 No. The model has no door swing. It would need to keep the swing.
6.9 The chair and gap are at different scales, so the sizes cannot be compared.
6.10 $5.9 \times 10 = 59$, $5.8 \times 10 = 58$, difference 1 cm.
6.11 "The 1 : 50 model could not read a 1 cm margin, and a 1 : 10 corner model fixed it."
6.12 Test against the real thing, use one scale, or name what was dropped.
6.13 Without a test, nobody knows which questions the model can answer.
6.14 to 6.16 Answers vary. Check that each circle's number is accounted for.
Slop check. (a) You cannot tell: no title, no labels. (b) Many colours, stars and joke text. (c) A title, a scale, labels for doors. (d) The stars and jokes.
Whole course. Answers vary. Sample: Book 2 kept one trial's result and set aside the other trials; Book 4 kept direction and size and set aside the cause of the push.

Words

diagram a drawing that shows how parts connect, not how they look.

gap the difference between what a model predicts and what is real.

map a flat drawing of a place that keeps places and distances.

margin the spare room a question has, such as a gap minus a chair's width.

model a simple stand-in that keeps some things and drops others.

prediction an answer written down before checking.

scale how big the model is compared with the real thing, such as 1 : 50.

scale bar a short ruler on a map that shows the scale.

Notes for grown-ups

This book teaches that a model is trustworthy for a question when its predictions about that question were compared with the real thing, and when it states which questions it cannot answer. Each chapter has one activity, one worked example with made-up practice numbers, one trust box, one "what did we keep?" prompt and a page of puzzles.

Materials and safety. A tape measure, a ruler marked in millimetres, paper, cardboard, scissors and tape. Use blunt scissors. Measure only rooms and paths that are safe to walk. Stay away from roads, stairs and ladders.

Scales. A scale such as 1 : 50 means every length on the model is the real length divided by 50. Children practise dividing and multiplying by 50, 20 and 10, and see that the choice of scale decides what can be shown.

The practice numbers were made up so the arithmetic is easy to check. Real rooms and hallways will give different numbers.

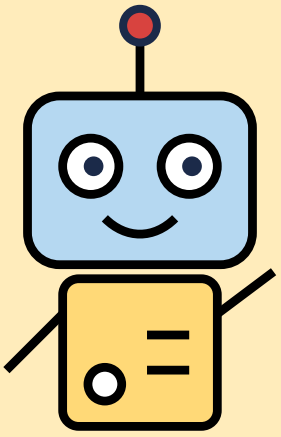
Sources

- E. R. Tufte. *Envisioning Information*. Graphics Press, 1990. For what a drawing keeps and what it drops.
- M. Monmonier. *How to Lie with Maps*, 3rd edition. University of Chicago Press, 2018. For scale and what maps leave out.
- J. R. Taylor. *An Introduction to Error Analysis*, 2nd edition. University Science Books, 1997. For measurement error and gaps.

That is the Science series

Nine books, one habit

Measure, test fairly, keep a ledger, sort with care and build models you check. Say what you kept and what you set aside. Take these habits into Math, Ecology, Spelling and Grammar, and Problem Solving.



Ages 9 to 11 ⋆ Science Book 9